Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Bring back an earlier idea

A planet has surface field 75 N/kg. At center distance 5 R, find field and the force on a 7 kg object.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

Your turn: Draw the force-time rectangle and shade its area.
Explain why the two impulse units are equivalent.

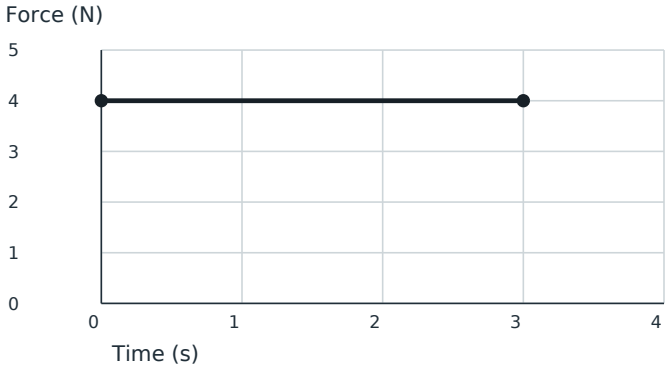
Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

Read the rectangular force-time graph.
Find the area and state what it means.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart starts at rest and receives 6 N·s rightward impulse. Final momentum is 6 kg·m/s.
Find final velocity by dividing by mass.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A spherical planet has surface field 144 N/kg.
Find the field at center distance 6 R, where R is the planet radius.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

07 Exit check

A 5 N force acts right for 2 s.
Find impulse. On a 2 kg cart initially at rest, find final velocity.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create two constant force-and-time pairs giving an impulse of 12 N·s.
Compare their force-time rectangles.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3